

Tutorial - Week 9

TG 11 - TG 12

1. Evaluate the following indefinite integrals using integration by parts.

(a) $\int x \sin x \, dx$

(e) $\int \sqrt{x} \ln x \, dx.$

(b) $\int x 2^{-x} \, dx.$

(f) $\int e^{\alpha x} \cos(\beta x) \, dx.$

(c) $\int x^2 e^x \, dx.$

(g) $\int \ln x \, dx.$

(d) $\int x \tan^{-1} x \, dx.$

(h) $\int \tan^{-1} x \, dx.$

2. Evaluate the integrals.

(a) $\int \sin^3 x \cos^2 x \, dx.$

(d) $\int \tan^3 x \sec^3 x \, dx.$

(b) $\int \sin^2 x \cos^3 x \, dx.$

(e) $\int \tan^4 x \sec^4 x \, dx.$

(c) $\int \sin^4 x \cos^4 x \, dx.$

(f) $\int \sin 7x \cos 5x \, dx.$

3. Integrate the following functions in at least two different ways.

(a) $\int_0^{\frac{\pi}{2}} \cos x \sin x \, dx.$

(c) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \tan^5 x \sec^6 x \, dx.$

(b) $\int_{-\pi}^{\pi} \sin^2 x \, dx.$

(d) $\int_0^{2\pi} \cos 2x \cos 3x \, dx.$

4. Find the integrals.

(a) $\int x^2 \sqrt{3 + 2x - x^2} \, dx.$

(c) $\int \frac{x^2}{(x^2 + a^2)^{\frac{3}{2}}} \, dx.$

(b) $\int \frac{1}{((ax)^2 - b^2)^{\frac{3}{2}}} \, dx.$

(d) $\int x \sqrt{1 - x^4} \, dx.$

5. Evaluate the following integrals of rational functions.

(a) $\int \frac{x}{(x+1)(x+2)(x+3)} \, dx.$

(g) $\int \frac{1}{x^2 + x + 2} \, dx.$

(b) $\int \frac{4x^2 - 15x + 15}{x^3 - 6x^2 + 11x - 6} \, dx.$

(h) $\int \frac{1}{(2x^2 + 1)(x - 1)} \, dx.$

(c) $\int \frac{2x^2 - 11x + 4}{x^2 - 5x + 6} \, dx.$

(i) $\int \frac{4x - 3}{3x^2 + 3x + 2} \, dx.$

(d) $\int \frac{x + 2}{(x + 1)^2} \, dx.$

(j) $\int \frac{3x^2 - 6x + 2}{x^3 - 2x^2 + x} \, dx.$

(e) $\int \frac{2x^2 + 2x - 1}{x^3 - 1} \, dx.$

(k) $\int \frac{3x^2 - 3x + 2}{x^3 - 2x^2 + 2x} \, dx.$

(f) $\int \frac{x - 1}{x^2 - x - 2} \, dx.$

(l) $\int \frac{4x^2 + 4x}{(x - 1)^2(x^2 + 1)^2} \, dx.$

6. Transform these integrals into integrals of rational functions and evaluate them.

(a) $\int \frac{1}{x\sqrt{1-x^2}} dx.$

(b) $\int \frac{1}{\sqrt{1+\sqrt[3]{x}}} dx.$

(c) $\int (x+1)^3 \sqrt{3-2x-x^2} dx.$

7. Consider the (inverse) change of variables $t = \tan\left(\frac{x}{2}\right)$. Then $\sin x = \frac{2t}{1+t^2}$ and $\cos x = \frac{1-t^2}{1+t^2}$. Using this, calculate the following indefinite integrals.

(a) $\int \frac{1}{\sin x + \cos x} dx.$

(b) $\int \frac{\sin x}{1 + \sin x + \cos x} dx.$